

Variance Reduction Techniques for Pricing Asian Options

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Options

- Options are financial derivatives that give the holder the right, but not the obligation, to buy or sell an underlying asset at a specified price within a specific time period.
- They are commonly used for hedging or speculating on the future price movements of assets such as stocks, commodities, or currencies.
- The payoff of a European call option is given by:

$$\phi(S) = (S(T) - K)^+$$

Pricing Options

- Risk-neutral valuation is used to price options:
 - ① Assume expected rate of return is the risk free rate.
 - ② Calculate the expected payoff of the option. For example, a European call option:

$$E[\text{payoff}] = E[(S(T) - K)^+]$$

- ③ Discount the expected payoff at the risk free rate. For the European call option:

$$C = e^{-rt} E[\text{payoff}]$$

Geometric Brownian Motion

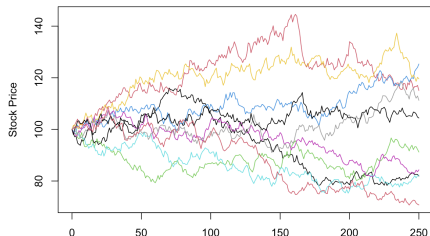
- Geometric Brownian motion is a continuous-time stochastic process used to model the price dynamics of financial assets.
- It is described by the stochastic differential equation:

$$dS = \mu S dt + \sigma S dW$$

where S is the asset price, μ is the drift, σ is the volatility, t is time, and dW is a Wiener process.



$$S_t = S_{t-1} e^{\left(\left(\mu - \frac{\sigma^2}{2}\right)\Delta t + \sigma Z \sqrt{\Delta t}\right)}$$



Asian Options

- The payoff for an Asian call option is:

$$\Phi(S) = (\bar{S} - K)^+$$

where

$$\bar{S} = \frac{1}{m} \sum_{i=1}^m S(t_i)$$

- The call price is:

$$C = e^{-rT} E \left[(\bar{S} - K)^+ \right]$$

which is estimated by:

$$\hat{C} = \frac{e^{-rt}}{n} \sum_{j=1}^n (\bar{S} - K)_j^+$$

Antithetic Variable Method

- The expected payoff is estimated by:

$$\hat{\theta}_{AV} = \frac{1}{\frac{n}{2}} \sum_{j=1}^{\frac{n}{2}} \frac{(\bar{S} - K)_j^+ + (\bar{S}' - K)_j^+}{2}$$

- The estimated price is given by:

$$\hat{C}_{AV} = e^{-rt} \left[\frac{1}{\frac{n}{2}} \sum_{j=1}^{\frac{n}{2}} \frac{(\bar{S} - K)_j^+ + (\bar{S}' - K)_j^+}{2} \right]$$

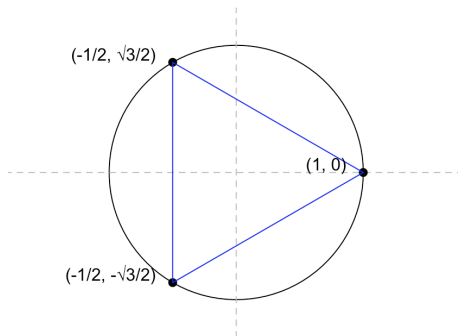
n-Simplex Method

- An object in geometry that generalises the idea of a triangle or tetrahedron to D-dimensions.



Figure: Simplices in different dimensions

Figure: 2-Simplex - Equilateral Triangle in Unit Circle



n-Simplex Method

Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{D+1}$ be the vertices of a regular D-simplex. The vertices have the following properties:

- $\|\mathbf{v}_i\|^2 = 1 \quad \forall i = 1, 2, \dots, D+1$ (by definition) and
- $\mathbf{v}_i' \mathbf{v}_j = -\frac{1}{D}$ for $i \neq j$

Let $\mathbf{Z} = \mathbf{Z}_1, \mathbf{Z}_2, \dots, \mathbf{Z}_D$ be a vector of D standard normal random variables, then:

- $\mathbf{v}_i' \mathbf{Z} \sim N(0, 1)$ and
- $\text{Cov}(\mathbf{v}_i' \mathbf{Z}, \mathbf{v}_j' \mathbf{Z}) = -\frac{1}{D}$

The D-simplex average of g is:

$$S(\mathbf{Z}) = \frac{1}{D+1} \sum_{i=1}^{D+1} g(\mathbf{v}_i' \mathbf{Z})$$

$\therefore$ the Monte Carlo estimator of $E[S(\mathbf{Z})]$ is:

$$\frac{1}{\frac{n}{D+1}} \sum_{j=1}^{\frac{n}{D+1}} S(\mathbf{Z}_j)$$

If g is monotone increasing, then:

- $\text{Cov}(g(\mathbf{v}'_1 \mathbf{Z}), g(\mathbf{v}'_2 \mathbf{Z})) \leq 0$
- $\therefore \text{Var}(S(\mathbf{Z})) \leq \frac{1}{D+1} \text{Var}(g(\mathbf{Z}))$

n-Simplex Method

- ① Compute $D+1$ vertices of the regular D -simplex: $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{D+1}$.
- ② Generate D vectors, $\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_D$, each containing m observations of standard normal random variables.

$$\mathbf{z}^{(1)} = z_1^{(1)}, z_2^{(1)}, \dots, z_m^{(1)}$$

$$\mathbf{z}^{(2)} = z_1^{(2)}, z_2^{(2)}, \dots, z_m^{(2)}$$

...

$$\mathbf{z}^{(D)} = z_1^{(D)}, z_2^{(D)}, \dots, z_m^{(D)}$$

n-Simplex Method

- ④ Generate $D+1$ stock price paths. Remember the model underlying the stock price behaviour:

$$s_{t_j}^{(1)} = s_{t_{j-1}} e^{((\mu - \frac{\sigma^2}{2})\Delta t + \sigma(x_{ij})\sqrt{\Delta t})}$$

$$s_{t_j}^{(2)} = s_{t_{j-1}} e^{((\mu - \frac{\sigma^2}{2})\Delta t + \sigma(x_{ij})\sqrt{\Delta t})}$$

...

$$s_{t_j}^{(D+1)} = s_{t_{j-1}} e^{((\mu - \frac{\sigma^2}{2})\Delta t + \sigma(x_{ij})\sqrt{\Delta t})}$$

for $j = 1, \dots, m$ and $i = 1, 2, \dots, D + 1$, where

$x_{ij} = [\mathbf{z}_j^{(1)}, \mathbf{z}_j^{(2)}, \dots, \mathbf{z}_j^{(D)}]' \mathbf{v}_i$ This generates the stock price path for m time steps.

- ⑤ Calculate the payoff from each path (there are $D+1$ paths) by summing up the m stock prices and dividing by m and then subtracting K from this result:

$$\text{Payoff} = (\bar{S} - K)^+ \text{ for } D+1 \text{ paths}$$

n-Simplex Method

- ⑥ Then calculate the average of the the $D+1$ payoffs:

$$S(\mathbf{z}) = \frac{1}{D+1} \sum_{i=1}^{D+1} \text{payoff}_i$$

- ⑦ Repeat steps 2 to 5 $\frac{n}{D+1}$ times.
- ⑧ Calculate the mean of the $\frac{n}{D+1}$ average payoffs.

$$\frac{1}{\frac{n}{D+1}} \sum_{j=1}^{\frac{n}{D+1}} S(\mathbf{z}_j)$$

- ⑨ Discount this using the risk-free rate to obtain the price of the Asian call option.

$$\frac{e^{-rt}}{\frac{n}{D+1}} \sum_{j=1}^{\frac{n}{D+1}} S(\mathbf{z}_j)$$

Results

s	r	σ	t	m
100	5%	20%	1	250

		Method									
		Direct		Antithetic		2-Simplex		3-Simplex		4-Simplex	
K	n	Price	Standard Error	Price	Standard Error	Price	Standard Error	Price	Standard Error	Price	Standard Error
90	1000	12.52	0.3230	12.57	0.0769	12.55	0.0564	12.57	0.0477	12.55	0.0430
	10000	12.44	0.1035	12.64	0.0267	12.63	0.0187	12.64	0.0161	12.62	0.0133
	100000	12.64	0.0329	12.62	0.0082	12.62	0.0059	12.61	0.0049	12.63	0.0043
100	1000	5.49	0.2456	6.21	0.1362	5.48	0.0860	5.71	0.0673	5.88	0.0596
	10000	5.76	0.0791	5.80	0.0395	5.77	0.0274	5.80	0.0227	5.86	0.0200
	100000	5.81	0.0254	5.81	0.0124	5.79	0.0087	5.79	0.0071	5.77	0.0062
110	1000	2.10	0.1571	2.01	0.0998	1.82	0.0765	2.03	0.0674	1.86	0.0517
	10000	2.00	0.0483	1.98	0.0315	2.05	0.0240	2.01	0.0197	2.04	0.0174
	100000	2.01	0.0156	1.98	0.0099	2.03	0.0076	2.01	0.0062	2.00	0.0055

Figure: Price and Standard Error

Results

		Method			
K	n	Antithetic	2-Simplex	3-Simplex	4-Simplex
90	1000	0.943	0.970	0.978	0.982
	10000	0.933	0.967	0.976	0.984
	100000	0.938	0.968	0.978	0.983
100	1000	0.692	0.877	0.925	0.941
	10000	0.751	0.880	0.918	0.936
	100000	0.761	0.883	0.922	0.940
110	1000	0.596	0.763	0.816	0.892
	10000	0.575	0.753	0.834	0.871
	100000	0.593	0.759	0.844	0.876

Figure: Variance Reduction Compared to Direct Method

Conclusions

- Variance reduction methods can improve the accuracy when pricing Asian options.
- n-simplex outperforms antithetic variables with higher dimensions, producing higher variance reduction.
- However, n-simplex is more complicated and harder to implement.